

$$T = 49 \text{ N}$$

9.

**Ans: D**

Between **Earth and Moon**,  $F_g = GM_e M_m / r^2$  --- (1)

Between **Sun and Earth**,  $170F_g = GM_e M_s / (400r)^2$  --- (2)

Solving (1) and (2),

$$M_s M_e = 2.7 \times 10^7$$

10.

**Ans: C**